Minimum Wage

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UBC

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- Plausibly exogenous change in price $\rightarrow$ what happens to quantity demanded.
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- Does it really?
- Minimum wage literature instrumental for (i) "Credibility Revolution" (ii) moving away from the perfectly competitive assumption of labor markets.

Introduction

- How do employers in low-wage labor markets respond to an increase in minimum wage?
- "Conventional" Economic Theory is unambiguous: cut employment (Stigler, 1946).
- Time series regressions (teenager employment on minimum wages).
- Aggregate data seems to confirm this prediction (Brown 1982, 1983; Wellington, 1991).

Wellington (1991)

Table 2
Estimated Effect of a 10 Percent Increase in the Minimum Wage on Teenage Employment by Race and Sex (in percent)[a] *1954–86*

Group	Linear	Logarithmic
1. Whites	−0.60	−0.64
	(1.25)	(1.34)
2. Nonwhites	−0.33	−0.62
	(0.37)	(0.75)
3. Males	−0.53	−0.84
	(1.05)	(1.68)
4. Females	−0.60	−0.35
	(0.99)	(0.63)

Opinion

The Right Minimum Wage: $0.00

Jan. 14, 1987


The New York Times

The Right Minimum Wage: $0.00

Howard Beach and the Governor

Piling On and the Cardinal

What the Riots in Soviet Asia Were About

See the article in its original context from January 14, 1987, Section A, Page 26 | Buy Reprints

The power of simplicity and transparency

Krueger: Yeah. I loved reading the *New England Journal of Medicine*.

Card: Yeah. And the *New England Journal* would come in every week, so there was a lot of stuff to read. And the beginning of each article would have "research design."

Krueger: And "methods."

Card: Yes, and if you've never seen that before and you were educated as an economist in the 1970s or 1980s, that just didn't make any sense. What is research design? And I remember one time I said, "I don't think my papers have a research design."

And so that whole set of terms entered economics as a result of those kind of changes in orientation.

Card and Krueger (1994)

- 410 fast-foods restaurants in New Jersey and Pennsylvania following the increase in NJ minimum wage from 4.25$ to 5.05$ per hours.
- Two research designs:
    - Difference in Differences (compare NJ and PA before and after law change).
    - Gap Design. (compare initially low wage establishments to high wage establishments before and after the design).
- The value of surveys and “zooming” in $\rightarrow$ employment, attrition, etc.

Two Waves

TABLE 1—SAMPLE DESIGN AND RESPONSE RATES

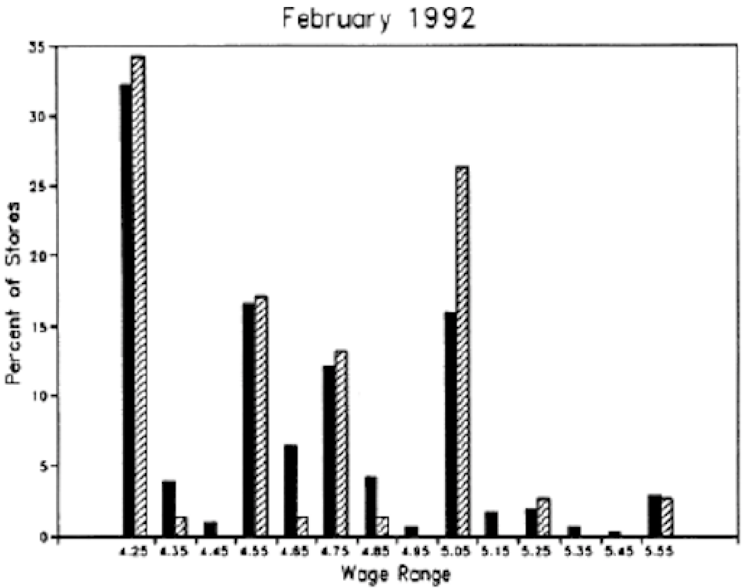
	All	Stores in: NJ	PA
Wave 1, February 15 – March 4, 1992:			
Number of stores in sample frame:[a]	473	364	109
Number of refusals:	63	33	30
Number interviewed:	410	331	79
Response rate (percentage):	86.7	90.9	72.5
Wave 2, November 5 – December 31, 1992:			
Number of stores in sample frame:	410	331	79
Number closed:	6	5	1
Number under rennovation:	2	2	0
Number temporarily closed:[b]	2	2	0
Number of refusals:	1	1	0
Number interviewed:[c]	399	321	78

[a] Stores with working phone numbers only; 29 stores in original sample frame had disconnected phone numbers.

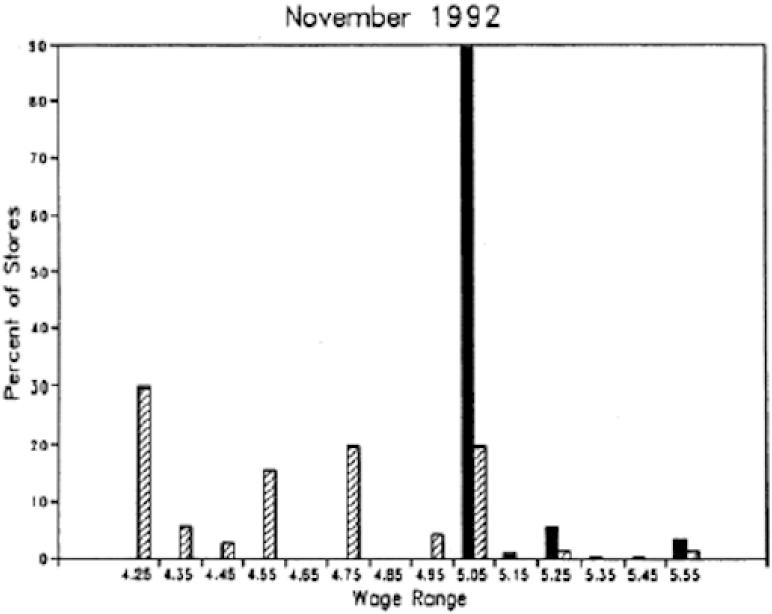
[b] Includes one store closed because of highway construction and one store closed because of a fire.

[c] Includes 371 phone interviews and 28 personal interviews of stores that refused an initial request for a phone interview.

Always make sure to measure your treatment reliably!



Ladies and Gentlemen we have a first stage!



PA fast food restaurants is a good control group?

TABLE 2—MEANS OF KEY VARIABLES

Variable	Stores in: NJ	Stores in: PA	t^a
1. *Distribution of Store Types (percentages):*			
a. Burger King	41.1	44.3	−0.5
b. KFC	20.5	15.2	1.2
c. Roy Rogers	24.8	21.5	0.6
d. Wendy's	13.6	19.0	−1.1
e. Company-owned	34.1	35.4	−0.2
2. *Means in Wave 1:*			
a. FTE employment	20.4	23.3	−2.0
	(0.51)	(1.35)	
b. Percentage full-time employees	32.8	35.0	−0.7
	(1.3)	(2.7)	
c. Starting wage	4.61	4.63	−0.4
	(0.02)	(0.04)	
d. Wage = $4.25 (percentage)	30.5	32.9	−0.4
	(2.5)	(5.3)	
e. Price of full meal	3.35	3.04	4.0
	(0.04)	(0.07)	
f. Hours open (weekday)	14.4	14.5	−0.3
	(0.2)	(0.3)	
g. Recruiting bonus	23.6	29.1	−1.0

Things look quite different in second wave (after law is passed)

3. *Means in Wave 2:*

a. FTE employment	21.0	21.2	−0.2
	(0.52)	(0.94)	
b. Percentage full-time employees	35.9	30.4	1.8
	(1.4)	(2.8)	
c. Starting wage	5.08	4.62	10.8
	(0.01)	(0.04)	
d. Wage = $4.25 (percentage)	0.0	25.3	—
		(4.9)	
e. Wage = $5.05 (percentage)	85.2	1.3	36.1
	(2.0)	(1.3)	
f. Price of full meal	3.41	3.03	5.0
	(0.04)	(0.07)	
g. Hours open (weekday)	14.4	14.7	−0.8
	(0.2)	(0.3)	
h. Recruiting bonus	20.3	23.4	−0.6
	(2.3)	(4.9)	

Notes: See text for definitions. Standard errors are given in parentheses.
[a]Test of equality of means in New Jersey and Pennsylvania.

Research Design #1

- Gap Design

$$\Delta E_i = a + bX_i + cGap_i + e_i \tag{1}$$

where

$$Gap_i = NJ_i \times \max\left(0, \frac{5.05 - W_{1i}}{W_{1i}}\right) \tag{2}$$

and W_{1i} is the reported starting wage of store i.

- Questions for you:

1. What is c measuring? (Mean value of $Gap_i = 0.10$).
2. Do I need PA fast-food restaurants to run this regression?
3. Is this a market level response?

Research Design #2

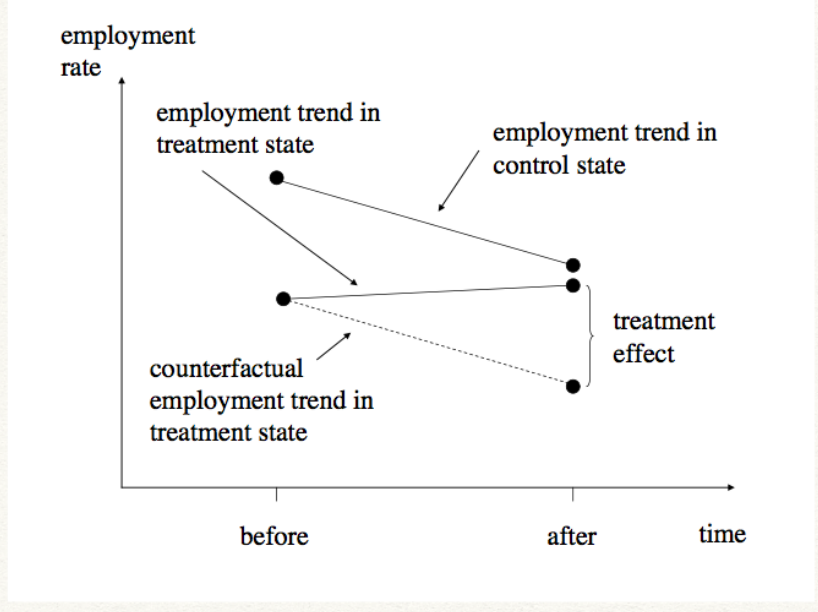
- Difference in Differences

$$E_{it} = \lambda_t + \gamma NJ_i + \delta(NJ_i \times Post_t) + r_{it} \tag{3}$$

- Card and Krueger estimate the model above in first differences (and adds controls).

1. Can you pin down δ from the previous table?
2. What is the key identifying assumption here? What can go wrong?
3. Is δ identifying a market level response?

Parallel Trends



Parallel Trends

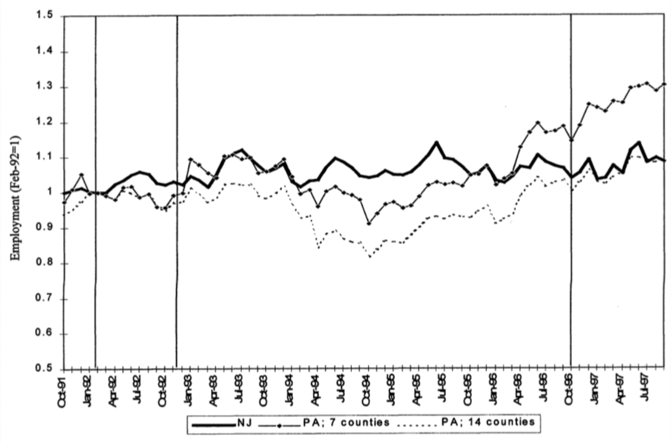


Figure 5.2.2: Employment in New Jersey and Pennsylvania fast-food restaurants, October 1991 to September 1997 (from Card and Krueger 2000). Vertical lines indicate dates of the original Card and Krueger (1994) survey and the October 1996 federal minimum-wage increase.

Zero or positive?

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

	Model				
Independent variable	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap[a]	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership[b]	no	yes	no	yes	yes
4. Controls for region[c]	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls[d]	—	0.34	—	0.44	0.40

Notes: Standard errors are given in parentheses. The sample consists of 357 stores with available data on employment and starting wages in waves 1 and 2. The dependent variable in all models is change in FTE employment. The mean and standard deviation of the dependent variable are −0.237 and 8.825, respectively. All models include an unrestricted constant (not reported).

[a]Proportional increase in starting wage necessary to raise starting wage to new minimum rate. For stores in Pennsylvania the wage gap is 0.

[b]Three dummy variables for chain type and whether or not the store is company-owned are included.

[c]Dummy variables for two regions of New Jersey and two regions of eastern Pennsylvania are included.

[d]Probability value of joint F test for exclusion of all control variables.

More expensive Big Macs?

TABLE 7—REDUCED-FORM MODELS FOR CHANGE IN THE PRICE OF A FULL MEAL

Independent variable	Dependent variable: change in the log price of a full meal				
	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	0.033 (0.014)	0.037 (0.014)	—	—	—
2. Initial wage gap[a]	—	—	0.077 (0.075)	0.146 (0.074)	0.063 (0.089)
3. Controls for chain and[b] ownership	no	yes	no	yes	yes
4. Controls for region[c]	no	no	no	no	yes
5. Standard error of regression	0.101	0.097	0.102	0.098	0.097

Notes: Standard errors are given in parentheses. Entries are estimated regression coefficients for models fit to the change in the log price of a full meal (entrée, medium soda, small fries). The sample contains 315 stores with valid data on prices, wages, and employment for waves 1 and 2. The mean and standard deviation of the dependent variable are 0.0173 and 0.1017, respectively.

[a]Proportional increase in starting wage necessary to raise the wage to the new minimum-wage rate. For stores in Pennsylvania the wage gap is 0.

[b]Three dummy variables for chain type and whether or not the store is company-owned are included.

[c]Dummy variables for two regions of New Jersey and two regions of eastern Pennsylvania are included.

Modern Version of This

Wages, Minimum Wages, and Price Pass-Through: The Case of McDonald's Restaurants
Orley C. Ashenfelter and Štěpán Jurajda
NBER Working Paper No. 28506
February 2021
JEL No. J23,J30,J38

ABSTRACT

We use price and wage data from McDonald's restaurants to provide evidence on wage increases, labor-saving technology introduction, and price pass-through by a large low-wage employer facing a flurry of minimum wage hikes from 2016-2020. We estimate an elasticity of hourly wage rates with respect to minimum wages of 0.7. In 40% of instances where minimum wages increase, McDonald's restaurants' wages are near the effective minimum wage level both before and after its increase; however, we also uncover a tendency among a large subset of restaurants to preserve their pay 'premium' above the minimum wage level. We find no association between the adoption of labor-saving touch screen ordering technology and minimum wage hikes. Our data imply that McDonald's restaurants pass through the higher costs of minimum wage increases in the form of higher prices of the Big Mac sandwich. We find a 0.2 price elasticity with respect to wage increases, which implies an elasticity of prices with respect to minimum wages of about 0.14. Based on a listing of all US McDonald's restaurants from 2010 to 2020, we also find no effects of minimum wages on McDonald's restaurant entry and exit.

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The power of zero

- A carefully thought out and transparent demonstration of the fragility of min wage results
  - Inferential issue: no clustering – but shouldn't affect GAP design
  - Results a bit under-powered to detect clear positive but strong enough to reject big negative
- Initial reaction of labor economists was not welcoming
  - Hamermesh (1995): employers anticipated the change in "before" period and "after" period too soon for adjustment to occur.
  - Neumark and Wascher (2000): results driven by measurement problems (responded to by Card and Krueger, 2000)

The Backlash

- Businessweek: "A Minimum Wage Study with Minimum Credibility"

  "Political correctness seems to have crept into the inner sanctum of the AEA, discrediting its scholarly journal and debasing its top prize. Unless the association cleans up its act, it can kiss its credibility goodbye"

- James Buchanan in the WSJ

  "Just as no physicist would claim that 'water runs uphill,' no self-respecting economist would claim that increases in the minimum wage increase employment. Such a claim, if seriously advanced, becomes equivalent to a denial that there is even minimal scientific content in economics, and that, in consequence, economists can do nothing but write as advocates for ideological interests."

- Merton Miller in the WSJ

  "Raising the minimum wage by law above its market determined equilibrium, they argue, actually costs nobody anything. (Or at worst, costs nobody very much because it's only a small, marginal increment, after all.) Is all this too good to be true? Damn right. But it sure plays well in the opinion polls. I tremble for my profession."

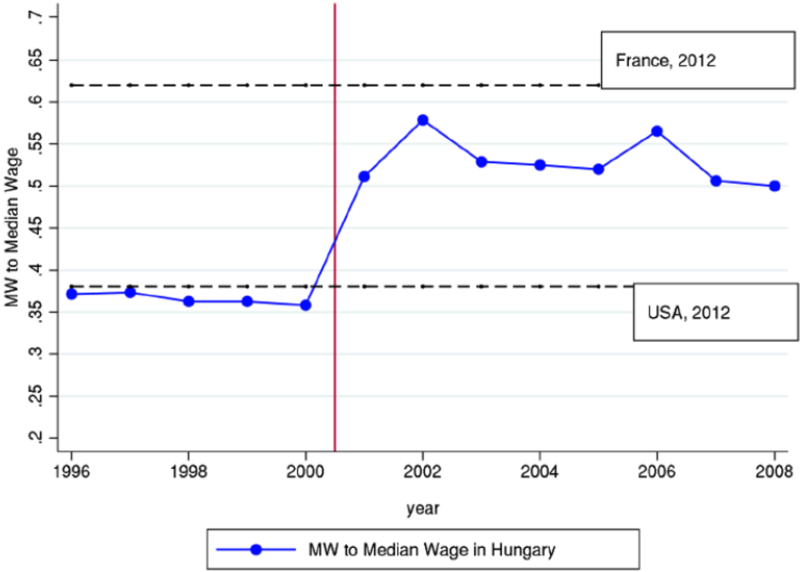
Interpretation

- What to make of these results?
  - Card-Krueger argue that positive employment effects reflect that not all workers are paid "the margin" but firms have latitude to set wages (like in monospony models).
- This is the key punchline of CK! Took many years to labor economists to digest this.
- Do GAP design and diff in diff identify the same parameter?
  - Diff in diff measures market-wide response
  - GAP measures effect of raising wage on a single firm holding market constant

Harasztosi and Lindner (2018)

- US min wage variation tends to be small and short run in nature
- Hungary experienced a large (60%) and persistent (˜8 years) increase in min wage in 2001

Now we're talking..



Harasztosi and Lindner (2018)

- One reform, one country: need a control group!
- Lesson learned from CK: use Gap design!

Harasztosi and Lindner (2018)

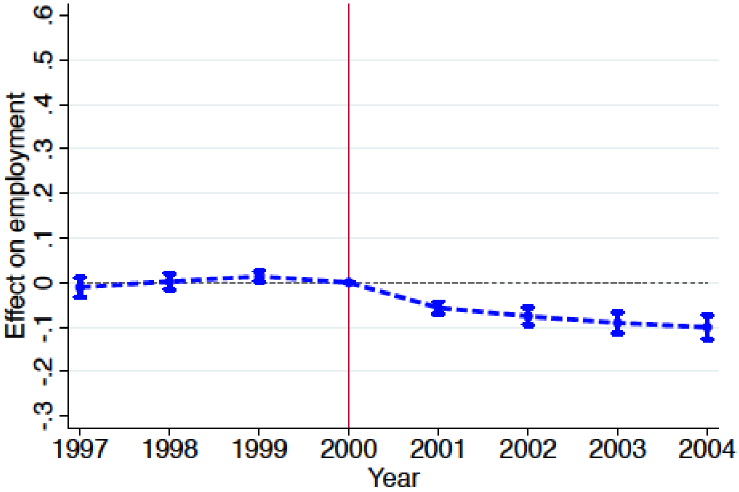
- One reform, one country: need a control group!
- Lesson learned from CK: use Gap design!
- Admin data from Hungary $\rightarrow$ can do Gap design on "steroids"
- Use firm level exposure design to infer MW effects

$$\frac{y_{it} - y_{i2000}}{y_{i2000}} = \lambda_t + \beta_t FA_i + r_{it} \tag{4}$$

where FA_i is fraction of workers affected in firm i, i.e. the share of workers in firm i being paid below the new minimum wage.

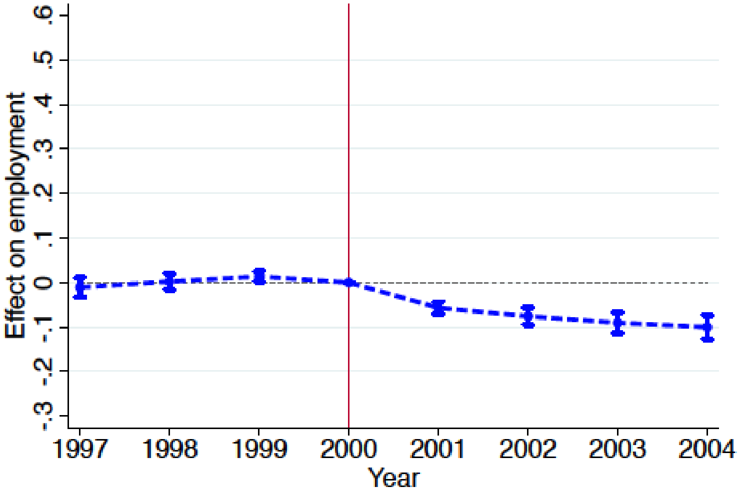
- What are the parametric assumptions of this model?
- What would do with dying firms?
- What would do with new firms?

Firm exposure lowers employment



(a) Effect on Employment

Is this effect large or small?



(a) Effect on Employment

Firm exposure increases wages and total labor costs

(b) Effect on Average Labor Cost

Can you back up an elasticity of labor demand from these graphs?

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Table 2: Employment and Wage Effects

	(1)	(2)	(3)	(4)	(5)	(6)
	Main Changes between 2000 and 2002		Main Changes between 2000 and 2004		Placebo Changes between 2000 and 1998	
Panel A: Change in Firm-Level Employment						
Fraction Affected	-0.078	-0.076	-0.093	-0.100	-0.003	0.002
	(0.008)	(0.010)	(0.012)	(0.012)	(0.008)	(0.009)
Constant	-0.050		-0.105		0.046	
	(0.005)		(0.007)		(0.005)	
Observations	19,485	19,485	19,485	19,485	19,485	19,485
Employment elasticity wrt. *MW* (directly affected)	-0.11	-0.10	-0.15	-0.15		
	(0.01)	(0.01)	(0.02)	(0.02)		
Panel B: Change in Firm-Level Average Wage						
Fraction Affected	0.53	0.58	0.48	0.54	-0.02	-0.03
	(0.01)	(0.01)	(0.01)	(0.01)	(0.003)	(0.01)
Constant	0.08		0.16		-0.08	
	(0.002)		(0.01)		(0.001)	
Observations	18,415	18,415	16,980	16,980	19,485	19,485
Employment elasticity wrt. wage	-0.15	-0.13	-0.20	-0.18		
	(0.02)	(0.02)	(0.03)	(0.03)		
Panel C: Change in Firm-Level Average Cost of Labor						
Fraction Affected	0.47	0.49	0.41	0.43	-0.03	-0.04
	(0.01)	(0.01)	(0.01)	(0.01)	(0.003)	(0.01)
Constant	0.04		0.10		-0.04	
	(0.001)		(0.002)		(0.001)	
Observations	18,415	18,415	16,980	16,980	19,485	19,485
Employment elasticity wrt. cost of labor	-0.17	-0.16	-0.22	-0.23		
	(0.02)	(0.02)	(0.03)	(0.03)		
Controls	no	yes	no	yes	no	yes

How do employers respond?

How do employers respond?

Table 3: Firms' Margins of Adjustment

	(1) Main Changes between 2000 and 2002	(2) Main Changes between 2000 and 2004	(3) Placebo Changes between 2000 and 1998
Panel A: Change in Total Labor Cost			
Fraction Affected	0.325	0.238	-0.031
	(0.013)	(0.020)	(0.009)
Panel B: Change in Revenue			
Fraction Affected	0.066	0.036	-0.020
	(0.013)	(0.018)	(0.015)
Panel C: Change in Materials			
Fraction Affected	0.049	0.021	-0.008
	(0.014)	(0.019)	(0.019)
Panel D: Change in Capital			
Fraction Affected	0.148	0.270	-0.006
	(0.034)	(0.054)	(0.015)
Panel E: Change in profits (relative to revenue in 2000)			
Fraction Affected	-0.011	-0.008	0.006
	(0.003)	(0.004)	(0.004)
Observations	19,485	19,485	19,485
Controls	yes	yes	yes

Higher Q or Higher P? Use data on Price indexes for Manufacturing firms

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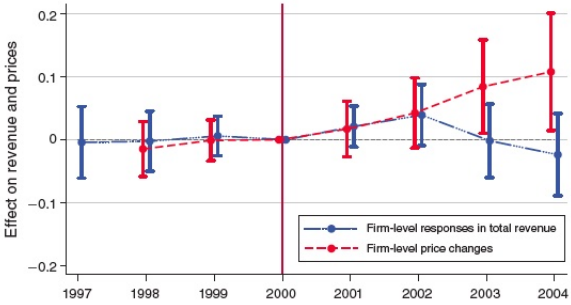
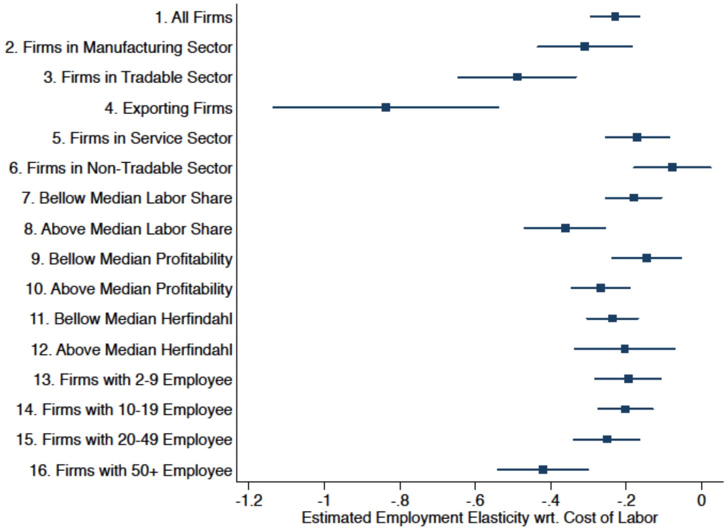


FIGURE 4. EFFECT ON PRICE INDEX AND REVENUE IN THE MANUFACTURING SECTOR

Notes: The figure shows the relationship between changes in different outcome variables and the fraction of workers affected by the minimum wage hike over time (beta coefficients with its 95 percent confidence intervals from equation (1)). The red dashed line shows firm-level price changes, while the blue solid line shows the revenue changes. The revenue changes include extensive (firm closure) and intensive margin responses. Since price data are available only for the manufacturing sector, we restrict our analysis to that sector. Controls are also included in the regressions.

How do you think this graph would look for other sectors (i.e. services)?

Employers in tradable sector responds differently...



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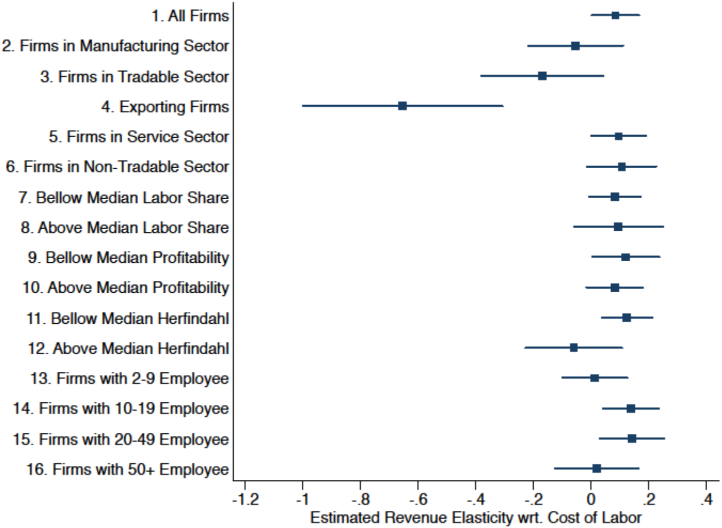

(b) Revenue Elasticity

Who pays for the minimum wage then?

Who pays for the minimum wage then?

by firm owners.

B. *Incidence of the Minimum Wage*

Our estimates above can be used to assess the incidence of the minimum wage
Our starting point is the following accounting identity:

$$Profit \equiv Revenue - Material - LaborCost - Depr - MiscItems,$$

where *Depr* is depreciation expenses, while *MiscItems* includes minor accounting
items (e.g., accrual deferrals). This equation leads to the following expression:[21]

$$(2) \qquad \frac{\Delta LaborCost}{Revenue_{2000}} = \underbrace{\frac{\Delta Revenue}{Revenue_{2000}} - \frac{\Delta Material}{Revenue_{2000}} - \frac{\Delta MiscItems}{Revenue_{2000}}}_{\text{Consumers Pay}} \underbrace{- \frac{\Delta Depr}{Revenue_{2000}} - \frac{\Delta Profit}{Revenue_{2000}}}_{\text{Firm Owners Pay}}.$$

Who pays for the minimum wage then?

Table 5: Incidence of the Minimum Wage

	(1) Changes between 2000 and 2002	(2) Changes between 2000 and 2004
Change in **Total Labor Cost** relative to Revenue in 2000	0.038	0.021
Change in Revenue relative to Revenue in 2000 ($\triangle Revanue$)	0.066	0.036
Change in Materials relative to Revenue in 2000 ($\triangle Material$)	0.033	0.014
Change in MiscItems relative to Revenue in 2000 ($\triangle MiscItems$)	0.005	0.005
Incidence on Consumers ($\triangle Revenue - \triangle Material - \triangle MiscItems$)	0.028	0.017
Change in Profits relative to Revenue in 2000 ($\triangle Profit$)	-0.011	-0.008
Change in Depreciation relative to Revenue in 2000 ($\triangle Depr$)	0.001	0.003
Incidence on firm-owners ($-\triangle Profit - \triangle Depr$)	0.010	0.005
Fraction paid by consumers (in percent)	74	77
Fraction paid by firm-owners (in percent)	26	23

Thoughts

- Negative but small employment elasticity $\rightarrow$ out of 300,000 min wage workers in Hungary 30K lost their job while 270K had their wage rise by 60%.
- Cost effects of min wage largely passed through to consumers!
- In an economy where min wage workers produced goods consumed by "rich" people, then MW effectively represents a redistribution from rich to poor.

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- Attila uses a neoclassical labod demand model (but with imperfect competition on the output market) to "fit" its reduced form effects.
  - Unclear how to distinguish effects of <u>firm</u>-specific from aggregate price shocks at the market level.
  - Monopsony model says small and big MW changes can have different signed effects

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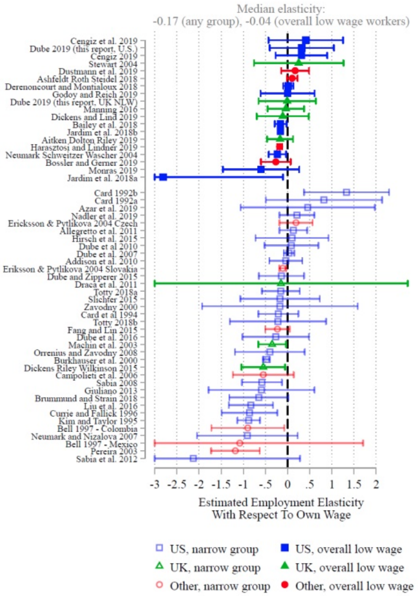
  - Unclear how to distinguish effects of firm-specific from aggregate price shocks at the market level.
  - Monopsony model says small and big MW changes can have different signed effects
- Policy implications
  - Price increases to domestic consumers are somewhat problematic
  - Who is consuming these goods?

Who pays the minimum wage?

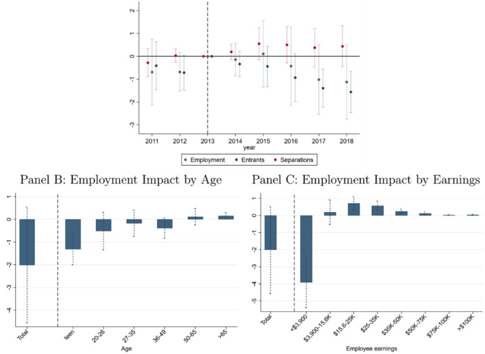
Chart 4.B: Own-wage employment elasticities from the minimum wage literature



More recent evidence from US Tax Returns of Independent Businesses: Wage Own Elasticity of around -0.25

Figure 2: Change in Employment Relationships Due to Minimum Wage Increases

Panel A: Change in Number of Employment Relationships



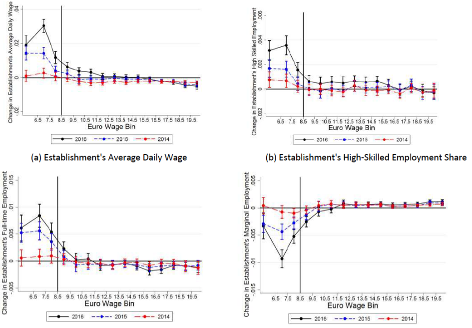
Note: The figure above describes the change in the number and composition of employment relationships due to the minimum wage. Our employment measure counts the number of W-2s attached to a firm each year and may change due to true changes in employment or changes in worker churn. Panel (a) traces how minimum wage increases affect the number of employment relationships of the average firm as well as the number of new (entrance) and incumbent (retention) workers. Entrance rates are estimated using Equation 1 where the outcome is the number of new hires, defined as employees of the firm in year t that were not with the firm in year t-1. Separation rates are estimated using the outcome of the number of separating workers, those at the firm in year t-1 that were not with the firm in year t. This decomposition is such that the coefficients from the entrants and separations specifications sum to the net employment response. Panels (b) and (c) decompose the 2013 to 2018 change in employment relationships by worker age and earnings in 2013.

Dustmann, Lindner, Schonberg, Umkehrer and vom Berge (2021)

- National wide introduction of the MW in Germany in 2015: 8.50 per hour (affecting around 15% of workers).
- High-quality matched employer-employee data
- Gap design at individual + region level (they have "high-quality" panel data)
- Null effect on employment. But large increase in wages for affected workers.
- Key new fact: "reallocation effect" of the minimum wage.

17% of the wage increase following MW due to Reallocation

Figure IV: Reallocation Effects of the Minimum Wage I: Individual Approach



(a) Establishment's Average Daily Wage

(b) Establishment's High-Skilled Employment Share

(c) Establishment's Full-Time Employment Share

(d) Establishment's Marginal Employment Share

Notes: The figure investigates the effect of the minimum wage on the reallocation of low-wage workers to establishments of higher quality. Establishment quality is measured before the minimum wage came into effect, and only changes for workers who switch establishments. In panel (a), we plot the change in the establishment's (log) average daily wage against the worker's initial wage bin for the periods 2012 vs 2014 to 2014 vs 2016, relative to 2011 vs 2013 pre-policy period, controlling for individual characteristics at baseline (age, education, sex, nationality, full-time status, district fixed effects and industry fixed effects). In panels (b) to (d), we repeat the analysis, using the change in the establishment's employment share of high-skilled workers (i.e., workers with a university degree); panel (b)) and the change in the establishment's employment share of full-time and marginally employed workers (workers who are employed in marginal part-time jobs) (panels (c) and (d)) as dependent variables. Estimates refer to coefficients δwt in regression equation (2). The black vertical line indicates the minimum wage of 8.50 Euro per hour. We also show the 95% confidence interval based on standard errors that are clustered at the district level.

What explains these positive re-allocation effects?

- Original motivation to introduce MW in New Zealand (1890): "helping worthwhile companies that employ working class breadwinners against sweatshops that gained market shares by undercutting those worthwhile companies"
- Same idea for Scandinavian model: Collective bargaining will drive low performing establishment out of the market.
- Underlying idea: firms have market power to set wages $\rightarrow$ wages depend upon productivity of the employer $\rightarrow$ MW kicks out "bad" firms.